HyperMIL: Adaptive Spatial–Semantic Hypergraph MIL for Survival Prediction from Whole-Slide Histopathology

Shenrong Chen, Xiangmin Han, Chenggang Yan, Yuhan Gao, and Zongpeng Li

***Abstract*—Survival prediction from whole-slide histopathology requires modeling both spatial tissue organization and morphological similarities among distant patches. Representing these complementary relations at multiple granularities while allowing patches to participate in more than one tissue context remains challenging. We propose HyperMIL, a spatial–semantic hypergraph multiple instance learning framework for patient-level survival prediction. Its structural construction recursively partitions a spatial–feature graph and retains both parent and child regions as overlapping hyperedges, preserving coarse and fine tissue contexts within a shared patch representation. Its semantic construction estimates slide-specific Gaussian-mixture prototypes and assigns patches to their two most similar prototypes through temperature-scaled cosine affinities, forming sparse, soft associations across spatially separated regions. Independent hypergraph neural networks propagate information within the two views, followed by separate gated-attention pooling and adaptive slide-level fusion. Averaging slide representations within each patient enables risk prediction using Cox partial likelihood. We evaluate HyperMIL on 1,291 patients from four cohorts of The Cancer Genome Atlas using patient-level five-fold cross-validation, complete slides, and common frozen patch features. HyperMIL achieves the highest observed macro concordance index of 0.7589 across 13 baselines and ranks first on three of the four cohorts. Six component ablations reduce performance across all four cohorts, supporting the complementary roles of overlapping spatial structure, soft semantic associations, and adaptive integration.**

***Index Terms*—computational pathology, hypergraph learning, multiple instance learning, survival analysis, whole-slide images**

I. Introduction

Whole-slide histopathology images (WSIs) preserve rich morphological and spatial information that can support prognostic modeling, but their gigapixel scale makes direct image-level learning difficult in routine survival pipelines. Multiple instance learning (MIL) therefore represents a WSI as a bag of patch embeddings and learns a slide- or patient-level predictor from weak supervision [1]–[4]. Although this formulation avoids dense regional annotation, prognostic evidence may depend on coordinated tissue patterns: spatially coherent regions, repeated feature patterns at distant locations, and interactions among multiple patches that are not naturally represented by independent instance scores.

Existing WSI MIL methods model such context in different ways. Attention-based models learn nonuniform instance weights, Transformer models introduce long-range token interactions, and graph models propagate information through pairwise neighborhoods [1], [4]–[6]. Hypergraph methods instead represent one relation with multiple patches and have been explored for WSI survival through multi-space slide hypergraphs, adaptive clustering, intra-/inter-patient computation, typed biological entities, and multimodal patient modeling [7]–[11]. These approaches differ substantially in both relation construction and input requirements. The specific question studied here is whether overlapping spatial regions at multiple granularities and soft feature-space memberships can provide complementary slide-local representations for patient-level survival prediction.

We address this setting with HyperMIL, a dual-hypergraph MIL framework for patient-level survival prediction. Multi-Granularity Adaptive Segmentation (MGAS) recursively partitions a spatial–feature graph and retains parent and child regions as overlapping structural hyperedges. Posterior-Guided Semantic Grouping (PASG) estimates slide-specific Gaussian-mixture prototypes and forms sparse weighted incidence from top-2 cosine affinities. Independent HGNN branches and gated-attention pooling preserve the two views until scalar WSI-level fusion. Multiple WSI embeddings are averaged within each patient before Cox–Efron risk prediction. Here, semantic denotes frozen feature-space organization rather than validated tissue phenotypes.

We evaluate HyperMIL under an input-matched patient-level protocol. All compared methods use complete WSIs and begin from the same frozen 512-dimensional patch-feature pool and the same patient folds, with spatial coordinates provided only to methods that require them. Direct spatial and feature-space GroupPool controls provide a reference for simple grouping and averaging, while six component variants examine the two branches, structural hierarchy, semantic assignments, fusion, and hypergraph propagation. Across four cohorts, HyperMIL obtains the highest observed macro C-index of 0.7589, with first-ranked means on three cohorts and a second-ranked mean on CHOL. Paired patient bootstrap

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analysis quantifies uncertainty in the comparisons.

The main contributions are summarized as follows:

- We introduce HyperMIL, a patient-level WSI survival framework that represents each slide through complementary structural and semantic hypergraphs, pools the two HGNN branches independently with gated attention, and combines their WSI-level representations through adaptive scalar late fusion.
- We develop two label-independent construction strategies with distinct inductive biases: MGAS retains accepted parent and child regions to form hierarchical overlapping structural incidence, whereas PASG uses regularized GMM-derived prototypes and top-2 temperature-scaled cosine affinities to form sparse overlapping semantic incidence.
- We perform an input-matched patient-level five-fold evaluation using common frozen patch features, complete-WSI baselines, direct grouping controls, and six component variants.

II. Related Work

A. Attention- and Transformer-Based WSI MIL

Attention-based MIL provides permutation-invariant bag aggregation while learning nonuniform instance weights. The gated formulation of Ilse *et al.* established a widely used attention-pooling baseline [1]. CLAM combines attention-based aggregation with instance-level clustering constraints for weakly supervised WSI analysis [2]. In computational pathology, DSMIL couples instance- and bag-level streams [3], DTFD-MIL uses pseudo-bags and two-tier feature distillation [12], ACMIL combines multiple attention branches with stochastic top-K instance masking [13], and AttriMIL introduces multi-branch attribute scoring with region- and slide-level constraints [14]. These methods enrich instance weighting and aggregation, but they do not explicitly represent one multi-patch relation through a hypergraph incidence structure.

Transformer-based MIL models dependencies directly among patch tokens. TransMIL uses Nyström self-attention together with positional encoding to capture correlated morphological and spatial information without constructing an explicit patch graph [5]. Token interaction and hypergraph propagation therefore encode context through different mechanisms: the former learns dependencies through attention, whereas the latter specifies multi-instance memberships through incidence and propagates information over the induced high-order relations.

B. Graph-Based Relational Modeling

Graph-based WSI models make relations explicit through pairwise edges. Patch-GCN represents a WSI as a spatial point cloud and propagates context over a coordinate-derived graph for survival prediction [4]. DualGCN-MIL constructs separate coordinate-space and feature-space graphs and fuses their representations for WSI classification [6]. TMEGL explicitly models interactions among tumor, lymphocyte, and stromal patches for survival prediction [15]. MorphoNet groups spatially adjacent, appearance-similar patches into sub-region embeddings before graph learning [16]. In contrast, MGAS retains patch vertices and represents nested regions as overlapping hyperedges. These graph models encode geometric or feature-space context through pairwise edges. Hypergraph incidence provides a complementary representation in which one structural region or feature-space group can connect multiple patches simultaneously; the empirical comparison determines how these alternatives perform in the studied cohorts.

C. Prototype-Based Slide Representation

PANTHER fits Gaussian mixtures to patch embeddings and concatenates mixture parameters into a compact, task-agnostic slide representation [17]. Thus, neither GMM-derived prototypes nor soft prototype assignments alone distinguish HyperMIL. PASG instead converts affinities to the fitted prototypes into a sparse incidence matrix for patch-level propagation; it does not use the concatenated mixture parameters as the slide embedding. Its top-2 cosine weights also differ from the posterior responsibilities used during mixture fitting.

D. Hypergraph-Based Survival Modeling

High-order WSI representations have also been studied directly for survival analysis. HGSurvNet models multilateral correlations in multiple representation spaces and introduces hypergraph max-mask convolution [7]. Hyper-AdaC constructs compact adaptive-clustering hypergraphs to model high-order relations among WSI regions [8]. IIHGC is particularly close at the construction level: its intra-WSI stage considers boundary-/spatial-wise topological hypergraphs together with a global-feature semantic hypergraph, followed by an inter-patient hypergraph stage for population-level modeling [9]. H2GT instead constructs a multi-scale heterogeneity-aware hypergraph over typed biological entities, such as cell clusters and tissue blocks [10]. HGMSurvNet uses a two-stage design in which patch-level WSI hypergraphs are followed by multimodal patient-level hypergraph learning that incorporates pathology, clinical records, and genomic profiles [11].

Recent work further expands this design space. HMHGT links hypergraphs at two image magnifications through vertical connections [18]; HyperMIL instead varies region granularity within one patch representation. ProSH-Net combines domain-adaptive pretraining with saliency-guided spatial and semantic hypergraphs [19], while DMCL-HFM pretrains hypergraph representations using masked reconstruction and contrastive learning [20]. TRIAGE-MIL combines tile selection with containment, intra-semantic, and inter-semantic hyperedges [21]. HyperMIL retains complete patch bags and separates recursive structural membership from prototype-based semantic incidence.

Within this family, HyperMIL focuses on the construction and combination of two slide-local relations: MGAS retains parent and child regions from recursive normalized-cut partitioning, PASG forms top-2 soft incidence from cosine affinities to slide-specific GMM prototypes, and the two branches interact only after independent WSI pooling. Table I

summarizes the corresponding construction-level differences with the closest hypergraph survival methods. HGMSurvNet is discussed qualitatively because its multimodal patient-level setting is outside the WSI-only configuration studied here.

III. Method

A. Problem Formulation and Patient-Level Survival Setting

Let $p \in \{1, \ldots, P\}$ index patients. Patient p may contribute $M_p \geq 1$ diagnostic whole-slide images (WSIs), but is associated with a single survival record (t_p, δ_p), where $t_p > 0$ denotes the observed time and $\delta_p \in \{0, 1\}$ is the event indicator ($\delta_p = 1$ for an observed event and $\delta_p = 0$ for right censoring). For the m-th WSI of patient p, the retained tissue region is represented as a bag of $N_{p,m}$ patch features and their two-dimensional spatial coordinates,

$$\mathcal{B}_{p,m} = \{(\mathbf{x}_{p,m,i}, \mathbf{q}_{p,m,i})\}_{i=1}^{N_{p,m}}, \tag{1}$$

where $\mathbf{x}_{p,m,i} \in \mathbb{R}^{512}$ denotes a frozen patch embedding and $\mathbf{q}_{p,m,i} \in \mathbb{R}^2$ denotes its spatial location on the WSI.

The learning objective is patient-level risk prediction from one or more WSIs,

$$r_p = f_\theta\Big(\{\mathcal{B}_{p,m}\}_{m=1}^{M_p}\Big), \tag{2}$$

where f_θ is parameterized by HyperMIL and $r_p \in \mathbb{R}$ is an unconstrained scalar risk score. Each patient contributes one risk value and one survival observation to optimization and evaluation.

B. WSI Preprocessing and Patch Representation

WSI preprocessing and patch-feature extraction are performed offline and remain fixed during survival-model training. Tissue regions are identified by Otsu-based foreground segmentation [22] and densely tiled from the level-0 WSI image into non-overlapping 256×256 patches. Each retained patch is encoded using an ImageNet-pretrained ResNet-34 feature extractor [23]. After removing the classification head, global spatial average pooling of the final convolutional feature map yields a 512-dimensional vector. The patch origin is retained in level-0 pixel coordinates, and the image encoder is not fine-tuned jointly with HyperMIL.

For a single WSI, we suppress the patient and slide indices and write

$$\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N]^\top \in \mathbb{R}^{N \times 512}, \tag{3}$$

$$\mathbf{Q} = [\mathbf{q}_1, \ldots, \mathbf{q}_N]^\top \in \mathbb{R}^{N \times 2}. \tag{4}$$

All eligible patches in the frozen WSI representation are retained for all compared methods; no patch subsampling or truncation is applied. The same frozen patch features are used across the comparison, with coordinates additionally provided to methods that require spatial information. Within HyperMIL, MGAS operates on both $\mathbf{X}$ and $\mathbf{Q}$, whereas PASG is constructed from $\mathbf{X}$ alone. The coordinates are converted to patch-grid units only inside MGAS, as described in Sec. III-D.

C. Overview of HyperMIL

For each WSI, MGAS constructs structural incidence $\mathbf{H}_S$ from $\mathbf{X}$ and $\mathbf{Q}$, whereas PASG constructs semantic incidence $\mathbf{H}_F$ from $\mathbf{X}$. Both constructions are slide-local, label-independent, and fixed during training. Independent HGNN encoders and gated-attention pooling yield branch representations for scalar late fusion. Fused WSI embeddings are averaged within each patient before risk prediction (Fig. 1).

D. Multi-Granularity Adaptive Segmentation

Multi-Granularity Adaptive Segmentation (MGAS) constructs a sparse structural hypergraph from local spatial–semantic connectivity. Because the stored WSI coordinates are expressed in pixels, we first rescale them by the patch size to obtain patch-grid coordinates,

$$\widetilde{\mathbf{q}}_i = \mathbf{q}_i / 256. \tag{5}$$

Let $\mathbf{A} \in \{0, 1\}^{N \times N}$ denote the adjacency matrix of the resulting undirected base graph. For $i \neq j$,

$$A_{ij} = \mathbb{I}\big(\|\widetilde{\mathbf{q}}_i - \widetilde{\mathbf{q}}_j\|_2 \leq \tau_s\big)\, \mathbb{I}\left(\frac{\mathbf{x}_i^\top \mathbf{x}_j}{\|\mathbf{x}_i\|_2 \|\mathbf{x}_j\|_2} \geq \tau_c\right), \tag{6}$$

with $A_{ii} = 0$. We use $\tau_s = 2.9$ in patch-grid units and $\tau_c = 0.75$. Thus, an edge is present only when two patches are both spatially close and sufficiently similar in the frozen feature space. The construction is slide-local and uses no survival labels.

MGAS processes each connected component of the base graph independently. Within a component C, it selects $c_C = \lfloor \sqrt{|C|} \rfloor$ highest-degree vertices as seeds and assigns the remaining vertices by round-robin multi-source breadth-first search, yielding spatially connected initial regions. Each initial region is then recursively considered for spectral bipartitioning. For a candidate region R, we form its normalized graph Laplacian and use the Fiedler vector to propose a binary split $R = A \cup B$, $A \cap B = \varnothing$, with the median Fiedler value as the partition threshold. The proposed split is evaluated by normalized cut [24],

$$\mathrm{Ncut}(A, B) = \frac{\mathrm{cut}(A, B)}{\mathrm{vol}(A)} + \frac{\mathrm{cut}(A, B)}{\mathrm{vol}(B)}, \tag{7}$$

where

$$\mathrm{cut}(A, B) = \sum_{i \in A} \sum_{j \in B} A_{ij}, \qquad \mathrm{vol}(A) = \sum_{i \in A} d_i, \tag{8}$$

and $d_i = \sum_{j \in R} A_{ij}$ is the degree of vertex i in the induced candidate-region graph. This criterion normalizes each cut by the corresponding induced-region volume instead of using the raw number of cut edges.

A candidate split is accepted only when both child regions satisfy the minimum-size constraint, $\mathrm{Ncut}(A, B) \leq \eta$, and the current recursion depth is below $D_{\max}$. In the final model, the minimum child size is two patches, $\eta = 0.8$, and $D_{\max} = 2$. Every initial region is retained as a structural hyperedge. When a split is accepted, its parent remains in the hyperedge set, both child regions are added, and recursion continues on the children until no admissible split remains or the maximum

TABLE I
CONSTRUCTION-LEVEL COMPARISON OF SELECTED ORIGINAL HYPERGRAPH SURVIVAL METHODS. NUMERICAL COMPARISONS USE THE ADAPTATIONS DESCRIBED IN SEC. IV-B.

Method	Construction / inference	Overlap / scale	Incidence	Representation / survival objective
HyperMIL	MGAS recursive normalized-cut hierarchy; PASG finite-GMM prototypes with top-2 cosine-affinity assignment	Parent–child structural hierarchy; top-2 semantic overlap	Binary MGAS; soft PASG	Branch-wise gated attention + scalar late fusion; Cox–Efron
HGSurvNet [7]	Multi-space WSI hypergraphs with hypergraph max-mask convolution	Multiple representation spaces	Hard/binary	Max-mask convolution; BCR
IIHGC [9]	Topological and semantic intra-WSI hypergraphs followed by an inter-patient hypergraph	Intra-WSI plus inter-patient levels	Hard/binary	Intra-/inter-risk fusion; Cox
Hyper-AdaC [8]	Adaptive clustering-based WSI hypergraph	Adaptive cluster scale	Hard/binary	Global pooling; Cox
H2GT [10]	Multi-scale hypergraph over typed biological entities	Multi-scale typed relations	Hard/binary	Attention pooling; discrete NLL

depth is reached. Consequently, ancestor and descendant hyperedges coexist, so a patch can be incident to multiple structural hyperedges at different granularities. Isolated vertices form singleton depth-0 hyperedges to preserve complete patch coverage. The resulting set $\mathcal{E}_S$ is therefore hierarchical and overlapping across levels rather than a mutually exclusive leaf partition.

The structural incidence matrix is binary,

$$H_S(i,e) = \begin{cases} 1, & v_i \in e, \\ 0, & \text{otherwise}, \end{cases} \qquad e \in \mathcal{E}_S. \tag{9}$$

MGAS therefore supplies the structural HGNN branch with adaptive patch groups derived from the same frozen patch features and coordinates used throughout the unified evaluation protocol.

E. Posterior-Guided Semantic Grouping

MGAS emphasizes spatially contiguous organization, whereas feature-similar tissue patterns may recur at distant locations within the same WSI. Posterior-Guided Semantic Grouping (PASG) therefore constructs a complementary weighted hypergraph in the frozen patch-feature space. PASG uses a regularized finite Gaussian mixture to discover slide-specific feature prototypes, but the final hypergraph incidence is defined by patch-to-prototype cosine affinity rather than by Gaussian-mixture posterior responsibility.

For each WSI, PASG fits a finite Gaussian mixture with at most K components,

$$p(\mathbf{x}_i) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \tag{10}$$

where π_k, $\boldsymbol{\mu}_k$, and $\boldsymbol{\Sigma}_k$ denote the mixture weight, mean, and full covariance of component k, respectively. The model is fitted by expectation–maximization (EM). Let γ_{ik} denote the posterior responsibility of component k for patch i,

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}, \tag{11}$$

and define the effective component count $N_k = \sum_i \gamma_{ik}$. The corresponding EM updates are

$$\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{i=1}^{N} \gamma_{ik} \mathbf{x}_i, \tag{12}$$

$$\boldsymbol{\Sigma}_k = \frac{1}{N_k} \sum_{i=1}^{N} \gamma_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top + \varepsilon \mathbf{I}, \tag{13}$$

$$\pi_k = \frac{N_k + \alpha}{N + K\alpha}. \tag{14}$$

In the final configuration, $K \le K_{\max} = 10$, $\alpha = 0.1$, and $\varepsilon = 10^{-6}$. Components with effective support $N_k < 2$ are discarded after fitting, leaving $K_{\text{eff}} \le K$ retained prototypes $\{\boldsymbol{\mu}_k\}_{k=1}^{K_{\text{eff}}}$. For numerical stability, near-empty components are reinitialized before the parameter update, covariance factorization uses adaptive diagonal jitter, and ill-conditioned fitting can fall back to double precision. Posterior responsibilities estimate and retain prototypes; cosine affinities define the incidence weights.

For each retained prototype, PASG computes cosine affinity in the frozen feature space,

$$s_{ik} = \frac{\mathbf{x}_i^\top \boldsymbol{\mu}_k}{\|\mathbf{x}_i\|_2 \, \|\boldsymbol{\mu}_k\|_2}. \tag{15}$$

Let

$$\mathcal{T}_i = \operatorname{Top}_{m_i}\left(\{s_{ik}\}_{k=1}^{K_{\text{eff}}}\right), \qquad m_i = \min(2, K_{\text{eff}}), \tag{16}$$

denote the retained prototype indices. The semantic incidence matrix $\mathbf{H}_F \in \mathbb{R}_+^{N \times K_{\text{eff}}}$ is defined by a temperature-scaled softmax over these affinities,

$$H_F(i,k) = \begin{cases} \dfrac{\exp(s_{ik}/\tau_F)}{\sum_{j \in \mathcal{T}_i} \exp(s_{ij}/\tau_F)}, & k \in \mathcal{T}_i, \\ 0, & \text{otherwise}, \end{cases} \tag{17}$$

where $\tau_F = 0.05$. Each patch is therefore incident to at most two semantic hyperedges, and its retained incidence weights sum to one. PASG consequently produces sparse, weighted, and overlapping semantic incidence: patches assigned to a common prototype need not be spatially adjacent, and a patch can contribute to two prototype-defined hyperedges with different strengths. The construction is performed independently for each WSI and uses no survival outcomes.

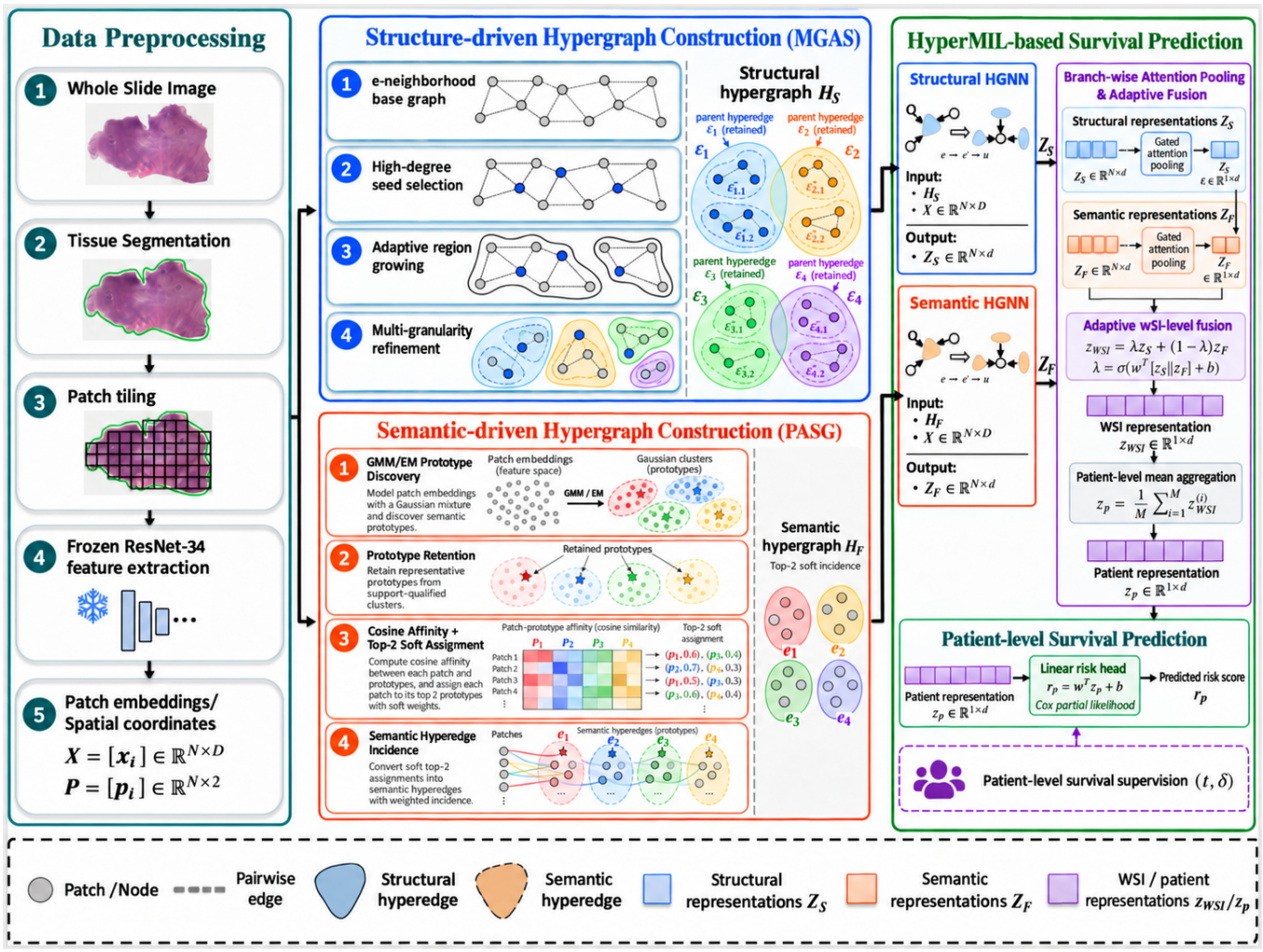


Fig. 1. HyperMIL workflow. Frozen ResNet-34 patch features support hierarchical structural hypergraphs (MGAS) and prototype-based soft semantic hypergraphs (PASG). Independent HGNN branches and gated-attention pooling produce WSI-level representations for scalar late fusion; arithmetic averaging across WSIs yields a patient representation for linear Cox risk prediction.

F. Dual-Branch Hypergraph Representation Learning

For either branch $b \in \{S, F\}$, let $\mathbf{H}_b \in \mathbb{R}_+^{N\times M_b}$ denote its incidence matrix. The corresponding hyperedge- and vertex-degree matrices are defined as

$$\mathbf{D}_{e,b}(e,e) = \sum_{i=1}^{N} H_b(i,e), \qquad \mathbf{D}_{v,b}(i,i) = \sum_{e=1}^{M_b} H_b(i,e). \tag{18}$$

Following the standard HGNN formulation [25], we employ the symmetrically normalized node–hyperedge–node propagation operator

$$\mathbf{P}_b = \mathbf{D}_{v,b}^{-\frac{1}{2}} \mathbf{H}_b \mathbf{D}_{e,b}^{-1} \mathbf{H}_b^{\top} \mathbf{D}_{v,b}^{-\frac{1}{2}}. \tag{19}$$

This formulation treats the entries of $\mathbf{H}_b$ as incidence strengths. Consequently, MGAS propagates information according to binary hierarchical region membership, whereas PASG propagates according to the relative cosine-affinity weights of its sparse soft semantic assignments.

Starting from $\mathbf{Z}_b^{(0)} = \mathbf{X}$, the ℓ-th layer of branch b is

$$\mathbf{Z}_b^{(\ell+1)} = \mathrm{Dropout}_{\rho}\Big[\mathrm{ReLU}\Big(\mathbf{P}_b \mathbf{Z}_b^{(\ell)} \mathbf{W}_b^{(\ell)} + \mathbf{1}(\boldsymbol{\beta}_b^{(\ell)})^{\top}\Big)\Big], \tag{20}$$

where $\mathbf{W}_b^{(\ell)}$ and $\boldsymbol{\beta}_b^{(\ell)}$ are branch-specific weights and biases, and $\rho = 0.2$ is the dropout probability. Each branch contains two layers with dimensions $512 \to 256 \to 256$. The structural and semantic branches have the same depth and hidden dimensionality but do not share learnable parameters. No residual connection or layer normalization is used. Their final patch representations are

$$\mathbf{Z}_S = \mathbf{Z}_S^{(2)}, \qquad \mathbf{Z}_F = \mathbf{Z}_F^{(2)}, \qquad \mathbf{Z}_S, \mathbf{Z}_F \in \mathbb{R}^{N\times 256}. \tag{21}$$

Although Eq. (19) is written in matrix form, HyperMIL evaluates the propagation through sparse incidence operations rather than explicitly materializing a dense $N \times N$ patch-to-patch matrix. This preserves the high-order aggregation induced by each hypergraph while keeping the representation-learning stage compatible with large WSI bags.

G. Branch-Wise Gated-Attention Pooling and Late Fusion

The structural and semantic HGNN branches are aggregated independently before they interact. For branch $b \in \{S, F\}$, let $\mathbf{z}_{b,i} \in \mathbb{R}^{256}$ denote the final HGNN embedding of patch i. A branch-specific gated-attention head computes

$$\mathbf{u}_{b,i} = \tanh(\mathbf{V}_b \mathbf{z}_{b,i} + \mathbf{c}_b) \odot \sigma(\mathbf{U}_b \mathbf{z}_{b,i} + \mathbf{d}_b), \tag{22}$$

$$e_{b,i} = \mathbf{w}_b^\top \mathbf{u}_{b,i} + b_{a,b}, \tag{23}$$

$$a_{b,i} = \frac{\exp(e_{b,i})}{\sum_{j=1}^{N} \exp(e_{b,j})}, \tag{24}$$

where σ denotes the sigmoid function and $\odot$ denotes element-wise multiplication. The projection weights and biases are learned independently for the two branches. The corresponding WSI-level branch representation is

$$\mathbf{z}_b = \sum_{i=1}^{N} a_{b,i} \mathbf{z}_{b,i}, \qquad \mathbf{z}_b \in \mathbb{R}^{256}. \tag{25}$$

The softmax is evaluated over all patches in the WSI, so each branch forms its own normalized patch-to-WSI aggregation according to its structural or semantic representation.

The two WSI-level branch representations are then combined by a learned scalar late-fusion gate,

$$\lambda = \sigma\big(\mathbf{w}_g^\top [\mathbf{z}_S \| \mathbf{z}_F] + b_g\big), \tag{26}$$

$$\mathbf{z}^{\mathrm{WSI}} = \lambda \mathbf{z}_S + (1 - \lambda)\mathbf{z}_F, \qquad \mathbf{z}^{\mathrm{WSI}} \in \mathbb{R}^{256}. \tag{27}$$

Here $\|$ denotes concatenation, and $\lambda \in (0, 1)$ is conditioned on the two branch-level WSI representations and adaptively balances structural and semantic evidence for the current slide. The two HGNN encoders, the two gated-attention heads, the scalar fusion gate, and the risk head are optimized jointly by the patient-level survival objective.

H. Patient-Level Risk Prediction and Cox–Efron Objective

For patient p with M_p diagnostic WSIs, HyperMIL first combines the WSI representations before risk prediction. Specifically, the patient representation is the arithmetic mean of the late-fused WSI embeddings,

$$\mathbf{z}_p = \frac{1}{M_p} \sum_{m=1}^{M_p} \mathbf{z}_{p,m}^{\mathrm{WSI}}, \qquad \mathbf{z}_p \in \mathbb{R}^{256}. \tag{28}$$

The patient representation is mapped to an unconstrained scalar log-risk score through a linear head,

$$r_p = \mathbf{w}_r^\top \mathbf{z}_p + b_r, \qquad \mathbf{w}_r \in \mathbb{R}^{256}, \quad b_r \in \mathbb{R}, \tag{29}$$

with no sigmoid or other probability-normalizing activation. A larger r_p therefore corresponds to a higher predicted relative hazard.

Training minimizes a patient-level Cox negative partial log-likelihood [26] evaluated over the complete training-fold risk sets. Let $\mathcal{P}_{\mathrm{tr}}$ denote the patients in the current training fold and let

$$E_{\mathrm{tr}} = \sum_{p \in \mathcal{P}_{\mathrm{tr}}} \delta_p \tag{30}$$

be the number of observed events. For the distinct event times $\{\tau_1, \ldots, \tau_J\}$, define

$$\mathcal{D}_j = \{p \in \mathcal{P}_{\mathrm{tr}} : t_p = \tau_j,\ \delta_p = 1\}, \qquad d_j = |\mathcal{D}_j|, \tag{31}$$

and the corresponding risk set

$$\mathcal{R}_j = \{p \in \mathcal{P}_{\mathrm{tr}} : t_p \geq \tau_j\}. \tag{32}$$

Tied event times are handled using Efron's correction [27]. The event-normalized negative partial log-likelihood is

$$\mathcal{L}_{\mathrm{Efron}} = -\frac{1}{E_{\mathrm{tr}}} \sum_{j=1}^{J} \left[\sum_{p \in \mathcal{D}_j} r_p - \sum_{\ell=0}^{d_j - 1} \log\left(\sum_{q \in \mathcal{R}_j} e^{r_q} - \frac{\ell}{d_j} \sum_{p \in \mathcal{D}_j} e^{r_p} \right) \right]. \tag{33}$$

All patient risks needed by Eq. (33) are assembled before the loss is evaluated. Hence, each Cox denominator is formed from the complete training-fold risk set rather than from an arbitrary mini-batch subset.

IV. Experiments

A. Datasets and Cohort Construction

We evaluated HyperMIL on four public cohorts from The Cancer Genome Atlas (TCGA) [28]: stomach adenocarcinoma (TCGA-STAD), thyroid carcinoma (TCGA-THCA), cholangiocarcinoma (TCGA-CHOL), and liver hepatocellular carcinoma (TCGA-LIHC). The survival endpoint is overall survival. Patients were included when at least one diagnostic WSI and a valid survival time/event indicator were available. The final study comprises 1,291 patients and 1,344 diagnostic WSIs (Table II). TCGA-THCA is strongly censored, with 16 observed events among 505 patients, whereas TCGA-CHOL contains only 39 patients with 19 observed events. We therefore interpret cohort-level results together with their fold-to-fold variability, particularly for these two cohorts.

All compared methods retain the complete frozen patch sets described in Sec. III-B, with bag sizes ranging from 15 to 167,524 patches per WSI (Table II). All diagnostic slides from each included patient are retained.

B. Compared Methods and Input-Matched Evaluation

For quantitative comparison, all compared methods use the same frozen 512-dimensional patch features and the same patient folds; spatial coordinates are additionally provided to methods that require them. The comparison includes MeanPool, ABMIL, PatchGCN, DSMIL, ACMIL, DTFD-MIL, TransMIL, AttriMIL, DualGCN-MIL, IIHGC, Hyper-AdaC, Spatial-KMeans-GroupPool, Semantic-KMeans-GroupPool, and HyperMIL. Method-specific attention, pseudo-bag construction, clustering, graph construction, and auxiliary regularization are retained when they are part of the original formulation.

For classification methods, we preserve their defining aggregation modules and adapt the final supervision to patient-level Cox–Efron survival prediction. Multi-slide patients use the same representation-level averaging policy described in

TABLE II
COHORT STATISTICS. PATCH COUNTS ARE REPORTED PER WSI AS MEDIAN [MINIMUM, MAXIMUM].

Cohort	Patients	WSIs	Events	Censored	Multi-slide patients	Patches/WSI
TCGA-STAD	387	413	163	224 (57.9%)	26	42,914 [15, 121,517]
TCGA-THCA	505	518	16	489 (96.8%)	12	54,620 [1,522, 167,524]
TCGA-CHOL	39	39	19	20 (51.3%)	0	79,570 [12,715, 138,126]
TCGA-LIHC	360	374	128	232 (64.4%)	13	53,147.5 [2,231, 141,351]

Sec. III-H. DSMIL retains its instance and bag streams, ACMIL retains multi-branch attention and its diversity regularization, DTFD-MIL retains pseudo-bag distillation and two-tier aggregation, TransMIL retains its Nyström-attention/PPEG architecture, and AttriMIL retains its two attribute branches and spatial constraint while omitting the classification-specific rank constraint. DualGCN-MIL is a paper-guided reimplementation using separate feature-space and coordinate-space graph branches.

IIHGC and Hyper-AdaC retain their respective hypergraph construction mechanisms. HGSurvNet is compared qualitatively, without a matched numerical evaluation. H2GT is also discussed qualitatively because it requires typed biological-entity inputs beyond the shared patch features and coordinates.

As simple grouping controls, we include Spatial-KMeans-GroupPool and Semantic-KMeans-GroupPool. Both use $K = \min(10, N)$, k-means++ initialization, 10 initializations, and random seed 42. Spatial K-means is applied to normalized two-dimensional coordinates, whereas Semantic K-means is applied to the frozen 512-dimensional patch features. Features are averaged within each nonempty group and then across groups to form the WSI representation, followed by the same multi-slide aggregation policy and a $512 \to 256 \to 1$ risk head.

C. Patient-Level Cross-Validation Protocol

All methods are evaluated using patient-level five-fold cross-validation within each cohort. All diagnostic WSIs from the same patient remain in the same fold, and models are trained for 100 epochs. The executed checkpoint-selection policy uses the highest held-out patient-level C-index; no separate inner validation is used.

All methods use patient-level Cox–Efron supervision with complete training-fold risk sets. For multi-slide patients, WSI embeddings are averaged before risk prediction, as defined in Sec. III-H.

D. Implementation Details

The MGAS and PASG construction parameters are specified in Secs. III-D and III-E. Gaussian-mixture fitting uses at most 100 iterations, convergence tolerance 10^{-3}, and random seed 42. Both hypergraphs are precomputed from frozen patch inputs and remain fixed during survival-model training. MGAS uses local spatial-neighbor search, and HGNN propagation uses sparse incidence operations.

HyperMIL is optimized with Adam [29] using a learning rate of 10^{-4}, weight decay 10^{-4}, and random seed 42; no learning-rate scheduler is used. An exact two-pass implementation first computes the complete-fold Cox loss and its derivatives with respect to patient risks, then replays each patient forward pass with identical dropout random states to accumulate parameter gradients. Adam performs one update per epoch without truncating WSI bags. Comparison methods use the same patient-level Cox–Efron supervision while retaining their method-specific optimization settings and auxiliary regularizers.

E. Evaluation Metrics

The primary performance measure is the patient-level Harrell concordance index (C-index) [30]. For each cohort and method in the main comparison and ablation study, we report the mean and sample standard deviation of the five fold-level C-indices. We additionally report the arithmetic mean of the four cohort-level mean C-indices as the cross-cohort macro average. For risk stratification, Kaplan–Meier curves [31] summarize groups defined by a within-fold median split of the out-of-fold predicted risks, with stratified log-rank tests [32] used to compare the survival distributions.

F. Paired Statistical Comparisons

We compare HyperMIL with each of the 13 baselines using $\Delta C = C_{\mathrm{HyperMIL}} - C_{\mathrm{baseline}}$. We obtain 95% confidence intervals from 10,000 paired patient bootstrap replicates [33] within each fold, stratified by event status. The same sampled patients are used for both methods in each replicate. Cohort-level differences average the five fold C-indices, and Macro is the equal-weight average across the four cohorts. The intervals are not adjusted for multiple comparisons. Macro differences and intervals are reported in Table V; complete cohort-wise estimates and intervals are provided in the accompanying machine-readable results. Resampling fixed predictions quantifies conditional patient-sampling uncertainty; it does not incorporate model retraining or remove checkpoint-selection bias.

G. Ablation Design

We compare the full pipeline (A0) with six variants. Structure-only (A1) and Semantic-only (A2) retain only the corresponding branch and its attention pooling. Leaf-only (A3) retains terminal structural regions and removes their ancestor hyperedges. Hard semantic (A4) assigns each patch exclusively to its highest-affinity retained prototype with unit incidence weight. Fixed fusion (A5) replaces the learned gate by $\lambda = 0.5$, averaging the two pooled branch representations. Incidence-based GroupPool (A6) replaces HGNN propagation with group pooling over the same MGAS and PASG incidence structures. All variants use the same complete WSI

TABLE III
PATIENT-LEVEL HARRELL C-INDEX (MEAN ± SAMPLE SD) UNDER FIVE-FOLD CROSS-VALIDATION USING COMPLETE WSIS. MACRO AVERAGES THE FOUR COHORTS. BEST AND SECOND-BEST MEANS ARE **BOLD** AND UNDERLINED, RANKED BEFORE ROUNDING.

Method	STAD	THCA	CHOL	LIHC	Macro
MeanPool	0.5630±0.0349	0.8726±0.0263	0.6597±0.1806	0.7164±0.0591	0.7029
ABMIL	0.5759±0.0632	0.8563±0.0565	0.6555±0.2126	0.7146±0.0762	0.7006
DSMIL	0.4965±0.0605	0.8296±0.0559	0.6557±0.2393	0.6932±0.0833	0.6688
ACMIL	0.5703±0.0621	0.8341±0.0657	0.7087±0.1855	0.7024±0.0702	0.7039
DTFD-MIL	0.5639±0.0581	0.8522±0.0719	0.7275±0.1376	0.7028±0.0833	0.7116
TransMIL	0.6086±0.0300	0.8848±0.0512	0.6667±0.1687	0.7131±0.0774	0.7183
AttriMIL†	0.5413±0.0377	0.8101±0.0639	0.6983±0.2791	0.7004±0.0701	0.6875
DualGCN-MIL‡	0.5693±0.0595	0.8844±0.0432	0.6212±0.1611	0.7217±0.0638	0.6991
PatchGCN	0.5867±0.0565	0.8974±0.0196	0.6992±0.2091	0.7216±0.0694	0.7262
Spatial-KMeans-GroupPool	0.5558±0.0366	0.8698±0.0278	0.6597±0.1806	0.7166±0.0612	0.7005
Semantic-KMeans-GroupPool	0.5589±0.0246	0.8628±0.0239	0.6735±0.2030	0.7152±0.0594	0.7026
IIHGC	0.5624±0.0364	0.8404±0.0467	**0.7471±0.0738**	0.7120±0.0841	0.7154
Hyper-AdaC	0.5663±0.0304	0.7946±0.1256	0.7315±0.1101	0.6214±0.0899	0.6784
HyperMIL (Ours)	**0.6200±0.0257**	**0.9310±0.0254**	0.7371±0.1202	**0.7473±0.0629**	**0.7589**

†AttriMIL omits its classification-specific rank constraint for survival prediction. ‡DualGCN-MIL is a paper-guided reimplementation because public model source code was unavailable.

inputs, frozen patch features, patient-level five-fold splits, training settings, and checkpoint-selection rule as A0; only the designated ablation is changed. The two single-branch variants also change model capacity. A hard single-prototype assignment changes both the incidence weights and the overlap of semantic memberships, so its comparison with soft top-2 incidence does not isolate weighting alone.

V. RESULTS

A. Main Comparison

Table III reports five-fold patient-level C-indices. HyperMIL ranks first by mean on STAD, THCA, and LIHC, and second on CHOL behind IIHGC (0.7371 versus 0.7471). It obtains the highest observed Macro of 0.7589, compared with 0.7262 for PatchGCN. Table V summarizes the paired Macro differences. The Macro difference from PatchGCN is $+0.0326$ (95% CI $[-0.0042, 0.0702]$); the interval includes zero despite the higher point estimate. Macro difference intervals lie above zero for 11 of the 13 baselines, with PatchGCN and DTFD-MIL as the exceptions; all CHOL difference intervals include zero.

B. Survival Risk Stratification

Figure 2 shows the survival distributions of the high- and low-risk groups. The reported stratified log-rank p-values are 6.59×10^{-7}, 4.35×10^{-4}, 0.005, and 1.34×10^{-10} for STAD, THCA, CHOL, and LIHC, respectively. High-risk groups have lower estimated survival probabilities, while uncertainty increases in the tails as the numbers at risk decline. These analyses describe within-cohort risk-group separation and do not test differences in predictive performance between methods.

C. Direct Spatial and Semantic Grouping Controls

Spatial-KMeans-GroupPool and Semantic-KMeans-GroupPool obtain macro C-indices of 0.7005 and 0.7026, versus 0.7029 for MeanPool and 0.7589 for HyperMIL.

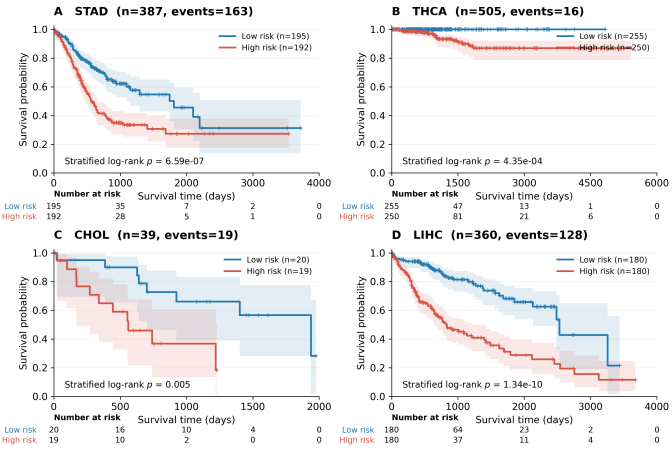


Fig. 2. HyperMIL Kaplan–Meier curves on (A) STAD, (B) THCA, (C) CHOL, and (D) LIHC, using out-of-fold predictions and within-fold median risk splits. Shading: 95% confidence intervals; ticks: censoring. Group sizes, stratified log-rank p-values, and numbers at risk are shown. Time is in days.

Neither grouping control improves the observed Macro over MeanPool. These are complete grouping controls, not isolated HyperMIL component ablations.

D. Ablation Study

Table IV reports the cohort means and sample standard deviations for the full pipeline and six variants. A0 obtains the highest reported mean on all four cohorts, with a macro C-index of 0.7589. Structure-only (A1) and Semantic-only (A2) obtain macro averages of 0.7068 and 0.7048, corresponding to decreases of 0.0520 and 0.0541 relative to A0. The full pipeline therefore exceeds either single-branch variant in the reported comparison.

Leaf-only (A3) obtains a macro average of 0.7139, compared with 0.7589 for A0. Hard semantic (A4) obtains 0.7082, with its largest cohort-level decrease on CHOL (0.6544 versus 0.7371). Fixed fusion (A5) obtains 0.7141, a macro decrease of 0.0447. Incidence-based GroupPool (A6) retains the same

TABLE IV
ABLATION STUDY USING PATIENT-LEVEL C-INDEX (MEAN ± SAMPLE SD) UNDER FIVE-FOLD CROSS-VALIDATION. MACRO IS THE ARITHMETIC MEAN ACROSS THE FOUR COHORTS. BOLD VALUES DENOTE THE HIGHEST MEAN IN EACH COLUMN.

Variant	TCGA-STAD	TCGA-THCA	TCGA-CHOL	TCGA-LIHC	Macro
A1: Structure-only	0.5738±0.0526	0.8594±0.0425	0.6899±0.1835	0.7043±0.0832	0.7068
A2: Semantic-only	0.5716±0.0365	0.8596±0.0554	0.6926±0.2143	0.6953±0.0602	0.7048
A3: Leaf-only	0.5751±0.0568	0.8694±0.0311	0.7056±0.1666	0.7055±0.0496	0.7139
A4: Hard semantic	0.5809±0.0532	0.8760±0.0314	0.6544±0.1236	0.7214±0.0572	0.7082
A5: Fixed fusion	0.5806±0.0437	0.8711±0.0290	0.6910±0.1436	0.7138±0.0200	0.7141
A6: Incidence-based GroupPool	0.5912±0.0582	0.8778±0.0307	0.6872±0.1628	0.7303±0.0703	0.7216
A0: Full pipeline	**0.6200±0.0257**	**0.9310±0.0254**	**0.7371±0.1202**	**0.7473±0.0629**	**0.7589**

TABLE V
PAIRED MACRO C-INDEX DIFFERENCES (Δ = HyperMIL − baseline) AND 95% CIS. POSITIVE VALUES FAVOR HYPERMIL.

Baseline	ΔMacro	95% CI
MeanPool	+0.0559	[0.0138, 0.1051]
ABMIL	+0.0583	[0.0171, 0.1110]
DSMIL	+0.0901	[0.0319, 0.1571]
ACMIL	+0.0550	[0.0019, 0.1211]
DTFD-MIL	+0.0473	[−0.0041, 0.1081]
TransMIL	+0.0406	[0.0042, 0.0879]
AttriMIL	+0.0713	[0.0199, 0.1209]
DualGCN-MIL	+0.0597	[0.0137, 0.1163]
PatchGCN	+0.0326	[−0.0042, 0.0702]
Spatial-KMeans-GroupPool	+0.0584	[0.0156, 0.1074]
Semantic-KMeans-GroupPool	+0.0563	[0.0096, 0.1067]
IIHGC	+0.0434	[0.0038, 0.1005]
Hyper-AdaC	+0.0804	[0.0119, 0.1598]

memberships but obtains a Macro of 0.7216, a decrease of 0.0373; A0 has higher observed means on all four cohorts. These summaries favor the full configuration but do not establish statistical significance. As noted in Sec. IV-G, the single-branch variants change model capacity, and hard assignment changes both weighting and membership overlap.

VI. DISCUSSION

A. Interpretation of Spatial–Semantic Modeling

MGAS represents nested spatial regions, whereas PASG connects distant patches through weighted prototype memberships. Independent propagation and pooling preserve these views until WSI-level fusion; the prototypes are feature-space groups, not validated tissue phenotypes. The full pipeline exceeds all component variants in observed cohort means, while the K-means grouping controls do not improve the macro result over MeanPool. A6 uses the same incidence structures and obtains a lower Macro than A0 (0.7216 versus 0.7589), favoring the HGNN configuration over direct group pooling. The controls address different questions: the K-means variants test simple grouping under alternative memberships, whereas A6 retains the proposed memberships and changes their processing. This distinction supports evaluating relation construction and propagation separately. Capacity and assignment changes still limit attribution to individual mechanisms. KM separation concerns risk groups rather than between-method performance.

B. Relation to Hypergraph Survival Models

HyperMIL shares the spatial–semantic motivation of IIHGC but retains slide-local computation rather than introducing inter-patient hypergraphs. Its hierarchical structural regions and prototype-based soft semantic incidence also differ from Hyper-AdaC's adaptive clustering and H2GT's typed multi-scale entities. Spatial–semantic relations, hierarchy, and prototypes are established design choices across these methods and more recent approaches [17]–[19], [21]. The contribution evaluated here is their particular realization as retained parent–child structural hyperedges, sparse cosine-weighted semantic incidence, and independent branch propagation with late fusion. The numerical comparison remains specific to the executed adaptations and shared feature representation.

C. Limitations and Future Directions

Evaluation is restricted to TCGA, without an independent external cohort. Selecting checkpoints by held-out fold performance can yield optimistic estimates; the reported results therefore require confirmation under independent model selection and external evaluation.

Survival information is uneven: THCA has 16 observed events among 505 patients, and CHOL has 39 patients with 19 events. Event scarcity and small cohort size limit the precision of cohort-level comparisons. Further evaluation on larger, independent multi-center cohorts is needed.

Using shared ImageNet-pretrained ResNet-34 features controls the input representation across methods, but performance with pathology-specific foundation encoders, such as UNI [34], remains untested. Contemporary prototype-based and pretrained hypergraph methods have not been evaluated under this protocol, so superiority over them is not established. The study also does not establish incremental value over clinical covariates. Mean WSI aggregation does not model between-slide heterogeneity. Paired uncertainty for component ablations remains unquantified; matched-capacity controls and separate tests of weighting and overlap would further clarify their contributions.

VII. CONCLUSION

HyperMIL combines overlapping hierarchical structural hypergraphs and sparse soft semantic hypergraphs through

independent branch pooling and scalar WSI-level fusion for patient-level Cox–Efron survival prediction. Its observed macro C-index is 0.7589, with first-ranked means on three cohorts and a second-ranked mean on CHOL. The full pipeline exceeds all six ablation variants in observed means. Independent cohorts, clinical adjustment, and more targeted component controls are needed to assess generalizability and attribution.

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